

ASSIGNMENT 1: MACHINE INFERENCE ATTACKS

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1 ASSIGNMENT

A resnet-18 model was given with Pub.pt file which contains some points which are members to the actual model and a pri.pt file for which we have to find the confidence scores, if these data points are in the actual model trained or not. In short, a Member Inference attack has to be carried on the model.

2 METHODOLOGY

This section will define the various approaches made for the member inference attacks conducted on the model

2.1 LOSS/CONFIDENCE/ENTROPY BASED ATTACK

These attacks did not require any input from public data, but relied on private data from priv.pt. Machine Learning models tend to be more confident when working on a data point on which it is being trained, which produces high confidence scores.

However, this approach was not very appropriate, as it did not use the public members, which could have been used to make the attack more elaborated. The score with this method was 0.050

2.2 LOGISTIC REGRESSION

The very first thing that can be used to predict scores between 0 and 1 would be Logistic Regression, which is a simple classifier. Basic idea was to train the logistic model with the public data and predict the private dataset.

However, this method did not work very well. This method was very slow as the model had to be trained epoch to epoch, even with a GPU. And after training, the classification model didn't work better than the above Loss/confidence/entropy model. The resulting scores always lied between 0.5 and 0.6. Normalizing the output and making shadow models did not improve the scoring.

2.3 XGBOOST CLASSIFIER

This method uses boosted trees to find a good gradient for classification. It is very fast and efficient. And it can also emulate properties of shadow models by creating n-tree, each of them carrying their own decisions while hiding some part of data from each tree. Processing thousands of trees was a matter of seconds unlike Logistic regression.

Starting with a very simple XGBoost classification model with only 300 trees, and each tree retaining 70 percent of information made score jump from 0.050 (from confidence method) to a good 0.052. Fine tuning the XGBoost parameters and using scale-pos- weight to balance classes improved the more. 'auc' worked better than other evaluation methods like 'logloss' and 'aucpr'. At the end the score was 0.059

2.4 CATBOOST CLASSIFIER

It is a XGBoost alternative, but using this with different parameters did not work as well as XGBoost as the score did not improve.

3 BASIC WORKING

Here is the basic Working of the Code:

- 1) Public and Private Features from pub.pt and priv.pt are extracted
- 2) Private features are extracted for prediction and Public features are extracted for training
- 3) XGBoost classifier trains on the model
- 4) XGBoost predicts the data membership with private features.
- 5) These predictions are saved in a CSV and submitted.

4 CONCLUSION

XGBoost works the best in this scenario. It is fast and can emulate the structure of a large shadow model for this attack. Other best classifier was CATboost, as it is a good alternative to XGBoost and performed better than other methods (except XGB). Confidence methods also seemed to work good as it was fast and did not need any model training from pub.pt. Logistic regression performed the worse, as it took lot of computational time and the score was worse than even a simple confidence method.